Probabilistic Static Voltage Stability Assessment for Distribution Network Considering Correlated Renewable Energy

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***Abstract*—In this paper, a probabilistic static voltage stability assessment method is proposed to study the influence of correlated renewable energy and loads on static voltage stability of distribution network. Through non-parametric kernel density estimation, the probability distribution function of distributed renewable generation output power is obtained, and the joint probability distribution is established by Copula function. Based on the quasi-monte Carlo simulation, the samples of random input variables are gained, and then the distribution of load margin is calculated. The efficiency and accuracy of quasi-monte Carlo simulation is verified, and the impact of the correlation between distributed generation on the static voltage stability of distribution network is studied. Besides, the influence of distributed generation configuration on voltage stability is also discussed.**

Keywords—distribution network*; *renewable generation*; *uncertainty*; *copula function*; *static voltage stability

I. INTRODUCTION

Recently, the penetration rate of distributed generation (DG) in the distribution network has gradually increased, where the wind power and photovoltaic power generation are mainly used[1]. However, due to the uncertainties and correlations of the output of DG and loads, the static voltage stability of the distribution network will be affected[2]. Therefore, in order to guarantee the stable operation of distribution network, it is necessary to study the probabilistic static voltage stability assessment method of the distribution network considering the uncertainties and correlations.

The static voltage stability analysis is based on the power flow equation. The load margin is obtained by calculating the critical point of the system static voltage stability. The common methods contain direct method, nonlinear programming method and continuous power flow method[3]. Further, considering the uncertainty in system, the scholars have proposed some probability assessment methods for static voltage stability. In [4], based on the point estimate method and the Cornish-Fisher series, a static voltage stability probability assessment method for the distribution network was proposed. In [3], through quasi-monte Carlo (QMC) simulation, a probabilistic static voltage stability method was proposed and the influence of the stochastic factors such as wind power and loads on the voltage stability margin is studied. In [5], combining the analytical method and approximation method, a probability model considering the random characteristics of the DG output was established. However, in above literatures, the DG output is obtained according to the parametric model of random variables, such as the Weibull distribution of wind speed and the Beta distribution of solar radiation intensity. The stochastic distribution characteristics in some regions cannot be simply described by the parametric model. Therefore, it is necessary to apply the non-parametric estimation method to depict the probability distribution of the random variable. Kernel density estimation (KDE) is a non-parametric estimation method which can gain the probability distribution of random input variables[6]. Meanwhile, there is a certain correlation between the same type of DG and the load, and their influences on the static voltage stability of the distribution network cannot be ignored.

In this paper, considering the uncertainties of DG output and loads, a probabilistic static voltage stability assessment method for distribution network is proposed. The probability distribution of DG output is generated by non-parametric KDE. The joint probability distribution between DGs with correlation is established by Copula function. Then, the samples of random input variables are obtained based on QMC simulation and the load margin is calculated by continuous power flow method. Through the simulation of IEEE 33-bus distribution network, the PDF of load margin is obtained. The efficiency and accuracy of QMC simulation are verified. Meanwhile, the effect of the correlation of DGs and loads on the static voltage stability is analyzed. The influence of DG configuration on improving voltage stability is further studied.

II. PROBABILISTIC STATIC VOLTAGE STABILITY

Continuous power flow is a method to analyze the system static voltage stability. In the distribution network, due to the uncertainty and correlation of DG output and loads, a probabilistic static voltage stability model is established based on the continuous power flow equation.

$$f(u,x)=0 \tag{1}$$

where u is the bus voltage; x is the random input variable.

In this paper, the random input variable x includes the DG outputs and load growth factor. Therefore, equation (1) is a

This research was supported by the Management Consulting Program of State Grid Corporation of China (Research on Energy Interconnection and Power Grid Operation Management Mechanism for Yangtze River Delta Integration, No. SGHD0000BGWT1900202).

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nonlinear equations containing random parameters, and its solution method is similar to the probabilistic load flow calculation[3]. Probabilistic load flow calculation methods mainly include Monte Carlo simulation (MCS) method, approximation method and analytical method. Compared with other methods, the Monte Carlo simulation method is robust, which can easily handle complex random problems. However, in order to obtain accurate results, MCS based on simple random sampling (SRS) requires a large amount of samples for repeated calculation which is time-consuming. Therefore, the QMC simulation method is introduced in this paper to reduce the computation burden. The generation process of low-difference Sobol sequence in QMC simulation is given in [7].

III. Uncertainty and Correlation of DG Output

This paper takes into account the uncertainty and correlation of DG output including wind power and photovoltaic power. Based on a large number of measurements, KDE is used to obtain the PDF of wind speed and solar radiation intensity. Considering the correlation of output of DG units, based on the Copula theory, a joint probability distribution model can be established. The samples of wind speed or solar radiation intensity are obtained according to the Copula function and the output power of DG units can be calculated.

A. Uncertainty of DG Output

Due to the uncertainty of the DG output, the parameter estimation method obtains the estimated values of the unknown parameters from the sample data based on the assumed PDF. However, when the random variable does not obey the previously assumed probability distribution, the method has a large error. KDE is a non-parametric method for estimating the probability distribution of random variables. The selection of kernel functions can adapt to the distribution characteristics of random variables.

Assuming that $x^1,x^2,\cdots,x^N$ are N samples of the random variable x, the KDE of the PDF of the random variable x is described as follows[6]

$$\hat{f}(x)=\frac{1}{Nh}\sum_{i=1}^{N}K\left(\frac{x-x^i}{h}\right) \tag{2}$$

where h is bandwidth; $K(\cdot)$ is Kernel function and satisfies the following conditions

$$\begin{cases} K(u)=K(-u) \\ \int K(u)du=1 \\ \int uK(u)du=0 \\ \int u^2K(u)du>0 \end{cases} \tag{3}$$

When $N\to\infty$, $h\to 0$ and $Nh\to\infty$, $\hat{f}$ is converged to f. In this paper, Gaussian distribution is used as a kernel function to estimate wind speed and solar radiation intensity. The mathematical expression of kernel function is given as

$$K(u)=\frac{1}{\sqrt{2\pi}}\exp(-\frac{1}{2}u^2) \tag{4}$$

The probability distribution of wind speed and solar radiation intensity can be obtained by KDE, and then the samples of DG output can be gained by QMC.

B. Correlation of DG Output

The Copula function is a joint function $F(x^1,x^2,\cdots,x^N)$ of the random variables $x^1,x^2,\cdots,x^N$ and their respective margin distribution functions, which is used to describe the correlation of random variables. The Copula function $C(u^1,u^2,\cdots,u^N)$ satisfies

$$F(x^1,x^2,\cdots,x^N)=C(F_1(x^1),F_2(x^2),\cdots F_n(x_n)) \tag{5}$$

Common Copula functions are Archimedes Copula function family and ellipsoid Copula function family. The Archimedes Copula function family includes Clayton-Copula, Gumbel-Copula and Frank-Copula functions. The ellipsoid Copula function includes Gaussian-Copula and t-Copula functions[8]. In this paper, the most suitable Copula function among the above five functions is used to construct the joint probability distribution function between different random variables by the K-S test.

Based on the above probability distribution or joint probability distribution function of the DG output, the Sobol sequence is used to obtain the wind speed or the solar radiation intensity. Then, the output power of the wind turbine or photovoltaic power plant can be calculated[5].

IV. Uncertainty and Correlation of Load

The uncertainty and correlation of the load are reflected in different growth modes of the load, depending on the load growth factor. It is assumed that there are m load buses in the distribution network, where the load types are mainly divided into residential load, commercial load, and industrial load. The correlation between the load growth factor depends on the type of load. In this paper, we first assume that the load growth factor $k_{L,i}$ satisfies the normal distribution.

$$f(k_{L,i})=\frac{1}{\sqrt{2\pi}\sigma_i}\exp(-\frac{(k_{L,i}-\mu_i)^2}{2\sigma_i^2}) \tag{6}$$

where μ_i and σ_i are the mean and standard deviation of the normal distribution, respectively.

The correlation coefficient matrix between the load growth factor can be expressed as

$$\rho=\begin{bmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1m} \\ \vdots & \vdots & \cdots & \vdots \\ \rho_{i1} & \rho_{i2} & \cdots & \rho_{im} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{m1} & \rho_{m2} & \cdots & \rho_{mm} \end{bmatrix} \tag{7}$$

where ρ_{ij} is the correlation coefficient between the load growth factor of nodes i and j ($0\le\rho_{ij}\le 1$, $\rho_{ii}=1$).

According to the QMC simulation for the probabilistic static voltage stability analysis, it is necessary to obtain the sample of the load growth factor as an input random variable. The sample of the load growth factor are generated as

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(1) Obtain the correlation coefficient matrix of load growth factor based on different load types ρ.
(2) Calculate the covariance matrix B of the load growth factor, where $B_{ij}=\sigma_i\sigma_j\rho_{ij}$.
(3) Perform Cholesky decomposition $B=LL^{\mathrm{T}}$ and obtain a lower triangular matrix L.
(4) Let $v_1, v_2,\ldots,v_m$ be multivariate independent standard normal distribution random variables, which is depicted as $(v_1, v_2,\ldots,v_m)\sim N(0, I)$. Set the sample size n and get sample matrix V for $(v_1, v_2,\ldots,v_m)$

$$V=\begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{m1} & v_{m2} & \cdots & v_{mn} \end{bmatrix} \tag{8}$$

where v_{ij} is the j-th sample of the i-th random variable v_i (i=1, 2,…,m, j=1, 2,…,n).

(5) The sample matrix K_{L} of the load growth factor ($k_{\mathrm{L},1}$, $k_{\mathrm{L},2},\ldots,k_{\mathrm{L},m}$) is:

$$K_{\mathrm{L}}=M+LV \tag{9}$$

where M is a matrix of m rows and n columns, and each column of M is the mean vector $\mu=[\mu_1, \mu_2,\ldots,\mu_m]^{\mathrm{T}}$ of ($k_{\mathrm{L},1}$, $k_{\mathrm{L},2},\ldots,k_{\mathrm{L},m}$).

V. Probabilistic Static Voltage Stability Assessment Procedure in Distribution Network

A procedure of probabilistic static voltage stability assessment is established in this paper based on the QMC simulation method. The uncertainty and correlation of DG output and load growth in distribution network are considered. The procedure is given as follows

(1) According to the measurements of the DG, obtain the probability distribution or joint probability distribution of the random input variables $y_1, y_2,\ldots,y_N$ by the KDE and Copula functions.
(2) Determine the sampling size n.
(3) Obtain a sample matrix Y based on the probability distribution or the joint probability distribution of the random variables $y_1,y_2,\ldots,y_N$ based on QMC simulation.
(4) Generate sample matrix K_{L} based on normal distribution parameters and correlation coefficient matrix of different load growth factors $k_{\mathrm{L},1}, k_{\mathrm{L},2},\ldots,k_{\mathrm{L},m}$.
(5) Generate the sample matrix Z of the DG output from the sample matrix Y based on the relationship between the DG output $z_1, z_2,\ldots,z_N$ and the random variables $y_1, y_2,\ldots,y_N$.
(6) Combine the sample matrices Z and K_{L} into the sample matrix X of the random input variable x in equation (1).
(7) d=1, let X_d be the d-th column of matrix X.
(8) Based on the random input variable X_d, calculate the load margin of the distribution network by continuous power flow.
(9) $d=d$+1, if $d>n$, proceed to the next calculation, otherwise go to (8);
(10) Obtain the probability distribution curve of load margin and analyze the operation risk of distribution network.

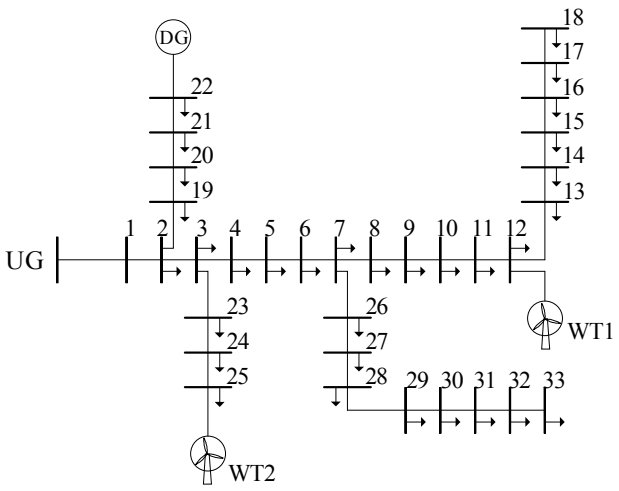


Fig. 1. Structure of 36-bus distribution network.

VI. Case Study

A. Data

In this paper, the IEEE 33-bus distribution network[9] is tested. The distribution network is connected to the utility grid through bus 1 and the reference voltage is 12.66kV. There are 4 DG units integrated in the distribution network including 2 wind turbines, 1 photovoltaic power plant and 1 gas turbine. The wind speed and the solar radiation intensity are collected within one year in a certain area of western China. The rated capacity of the wind turbine is 280kW. The cut-in, rated and cut-out wind speed of wind turbine are 3m/s, 15m/s and 20m/s, respectively. The area of the photovoltaic array is 2000m^2 and the photoelectric conversion efficiency is 14%. The load growth factor satisfies the normal distribution with the mean of 2.5 and the standard deviation of 0.25. The correlation coefficient between the same type of load is 0.8 and these between the different types of load is 0.2, which indicates the correlation.

In this paper, four cases are studied. Case 1: the random input variables are independent. Case 2: the correlation among loads is considered, regardless of the correlation of renewable energy. Case 3: the correlation of renewable energy is considered, regardless of the correlation between loads. Case 4: the correlation of renewable energy and loads are both considered.

TABLE I. VALUE OF d FOR FIVE COPULA FUNCTIONS

	Normal	t	Gumbel	Clayton	Frank
d	0.0341	0.0211	0.0182	0.0796	0.0273

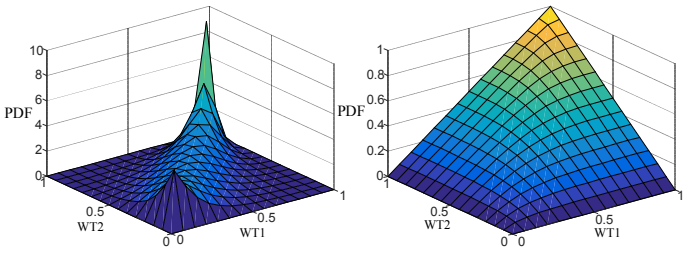


Fig. 2. Joint probability distribution of wind speed.

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TABLE II. EXPECTATION, STANDARD DEVIATION AND CALCULATION TIME OF LOAD MARGIN

Sampling mode	E/kW	Error	σ/kW	Error	Time/s
Reference	9788.4	—	450.6	—	28972
SRS	9828.9	0.41%	437.9	2.82%	330
Sobol	9796.7	0.08%	457.2	1.46%	323

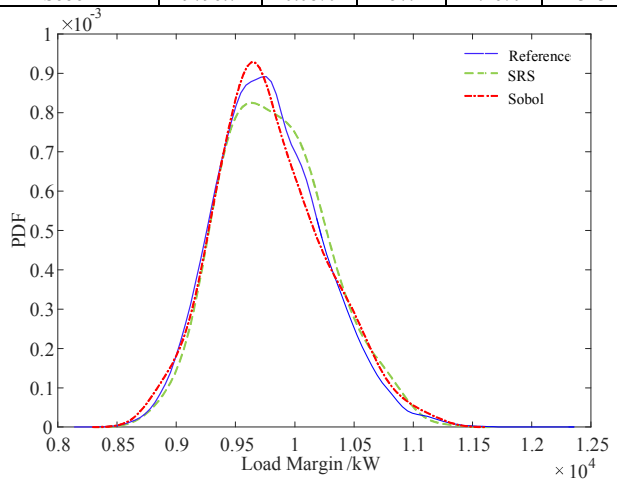


Fig. 3. PDF of load margin.

B. Comparison of sampling methods

The distribution network with 2 wind turbines and 1 gas turbine is illustrated in Fig. 1. In case 4, the Copula function is used to establish the joint probability distribution function of wind speed. Through the K-S test method, the most suitable Copula function is selected to describe the joint probability distribution between wind turbines. The infinite norm between the five common Copula functions and the empirical distribution function in the K-S test is represented as $d=\|\cdot\|_\infty$, as shown in Table 1.

By comparison, the Gumbel-Copula function with the best fitting degree is selected to describe the joint probability distribution between wind speeds. The joint probability density function and joint cumulative probability distribution function between wind turbine 1 and 2 are shown in Fig. 2.

Based on the joint probability distribution between the wind speeds, 500 wind speed samples are obtained by SRS method and Sobol sequences, respectively. The load margin is calculated using continuous power flow. The computation time for different sampling methods, the mean and standard deviation σ of the load margin are shown in Table 2. Meanwhile, the PDFs of the load margin is shown in Fig. 3.

According to the Table 2 and Fig. 3, the results of QMC simulation based on Sobol sequence under the same sampling times are closer to the reference value. With the same sample size, the QMC based on Sobol sequence can obtain more accurate results compared with SRS method. Therefore, the QMC based on Sobol sequence is utilized to perform a probabilistic static voltage stability assessment.

C. Correlation analysis

Based on the four designed cases in this paper, the PDFs and CDFs of the load margin are obtained as shown in Fig. 4 and Fig. 5.

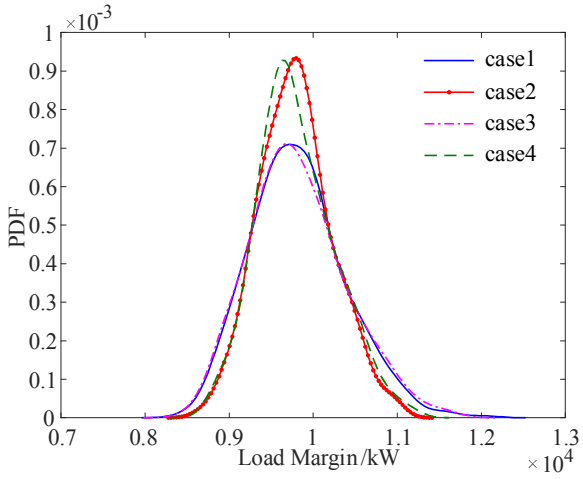


Fig. 4. PDF of load margin in different cases

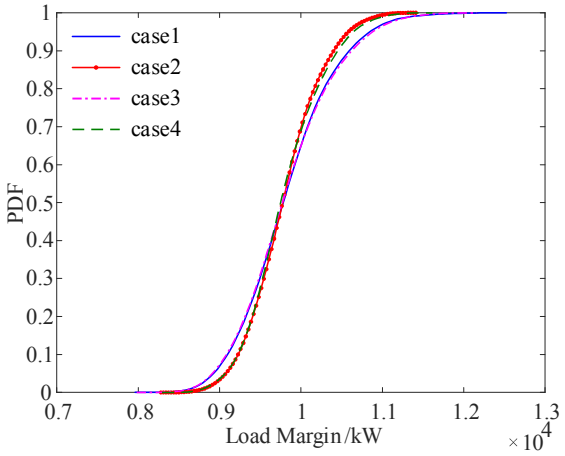


Fig. 5. CDF of load margin in different cases

According to the PDFs and CDFs in Fig. 4 and Fig. 5, it can be seen that the correlation among loads has a great influence on the probability distribution of the load margin. The correlation mainly affects the variance of the PDF of load margin. Meanwhile, the influence of correlation among wind power on the probability distribution of load margin is much smaller than that among loads. The reason is that the proportion of wind power output in the distribution network system is relatively small in this paper, and thus, its correlation has a relatively small impact on the probability distribution of the load margin.

D. DG configuration analysis

In the 33-bus distribution network system of Fig. 1, a photovoltaic power plant is integrated in this network with three modes. Meanwhile, the output power of photovoltaic cell and the wind turbine are independent. Taking into account the uncertainty of the solar radiation intensity the three modes are given as follows, mode 1: the photovoltaic cell does not access to the distribution network; mode 2: the photovoltaic cell is

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accessed to the distribution network at bus 8; mode 3: the photovoltaic cell is accessed to the distribution network at bus 18. The PDFs and CDFs of the load margin are obtained respectively, as shown in Fig. 6 and Fig. 7. Moreover, the mean and standard deviation of the load margins with the three modes are shown in Table 3.

From Fig. 6 and Fig. 7, it can be seen that compared with the mode 1, the PDFs and CDFs of mode 2 and mode 3 are both shifted to the right. Further, combined with Table 3, the mean of load margin is improved and the static voltage stability is enhanced in the distribution network. Meanwhile, compared the mode 2 and mode 3, it can be seen that the mean of the load margin is more greatly increased in mode 3. Therefore, the connection location of the DG units which is selected reasonably can further enhance the voltage stability of the distribution network.

TABLE III. MEAN AND STANDARD DEVIATION OF LOAD MARGIN

Mode	Mean/kW	σ/kW
Mode 1	9796.7	457.2
Mode 2	9876.7	483.8
Mode 3	10128.2	677.3

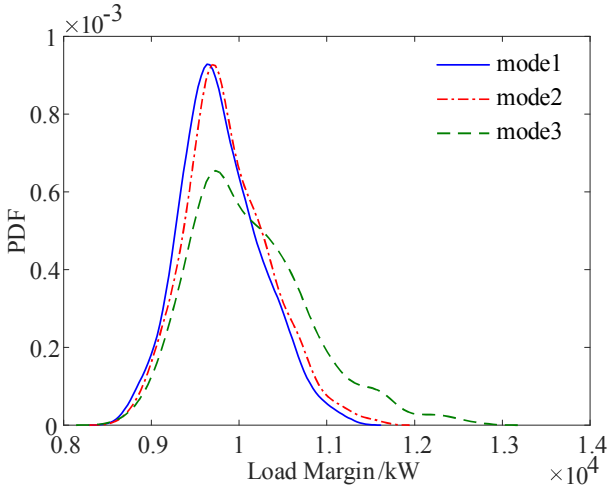


Fig. 6. PDF of load margin in different connection modes

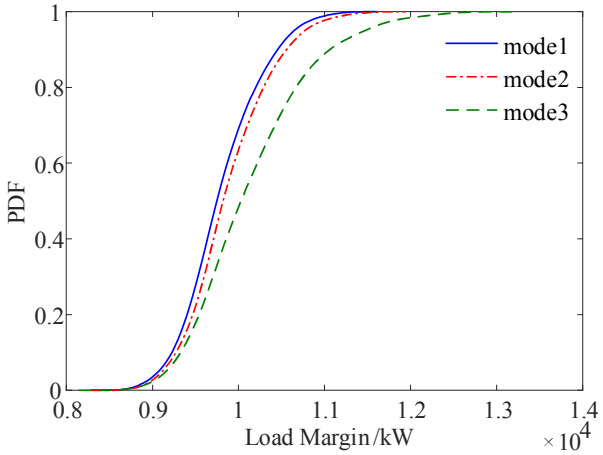


Fig. 7. CDF of load margin in different connection modes

VII. CONCLUSION

In this paper, considering the uncertainty and correlation of renewable energy and loads, a probabilistic static voltage stability assessment method for distribution network is proposed. The probability distribution of DG output power is obtained by non-parametric KDE method. Meanwhile, the Copula function is used to establish the joint probability distribution for the correlated DG output power. The samples of random input variables are obtained based on QMC simulation, and the load margin is calculated by continuous power flow. Through the simulation of the distribution network system, the main conclusions are given as follows:

(1) The QMC simulation based on Sobol sequence can improve the calculation efficiency and guarantee accuracy of results. Compared with the SRS-based MCS, it is more suitable for the probabilistic static voltage stability assessment of the distribution network.
(2) The correlation among DG and loads has a significant impact on the static voltage stability of the distribution network.
(3) The reasonable configuration for DG units is beneficial to improve the static voltage stability of system.

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